

Computer Architecture Introduction

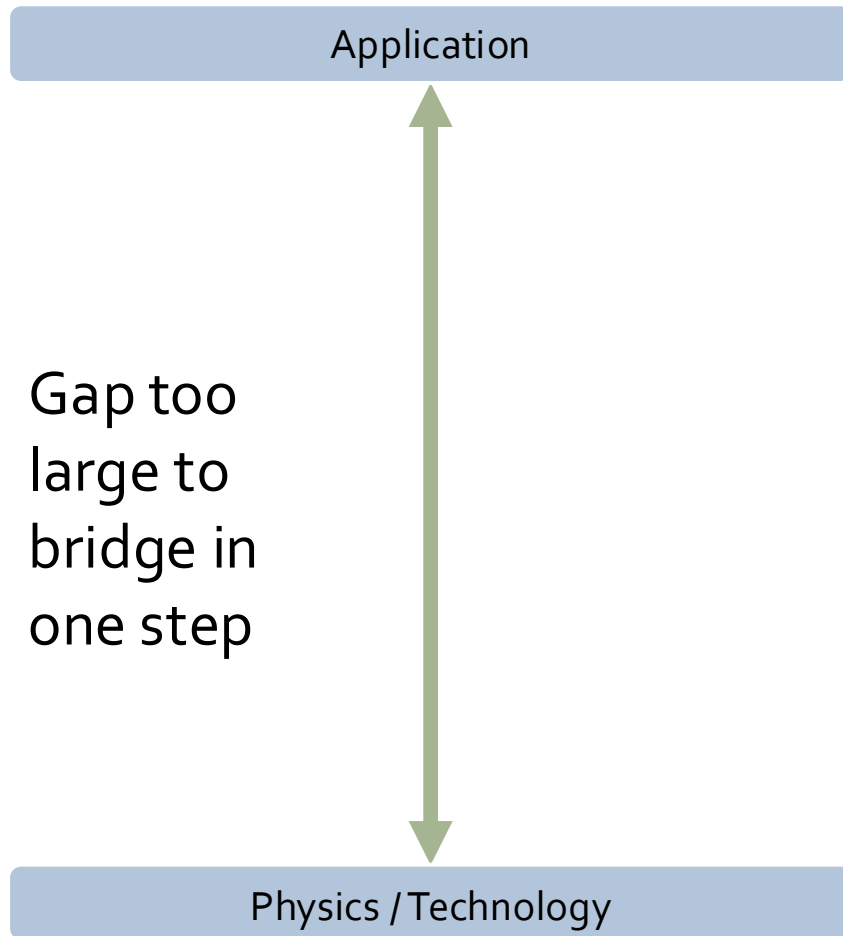
Ting-Jung Chang

NYCU CS

Agenda

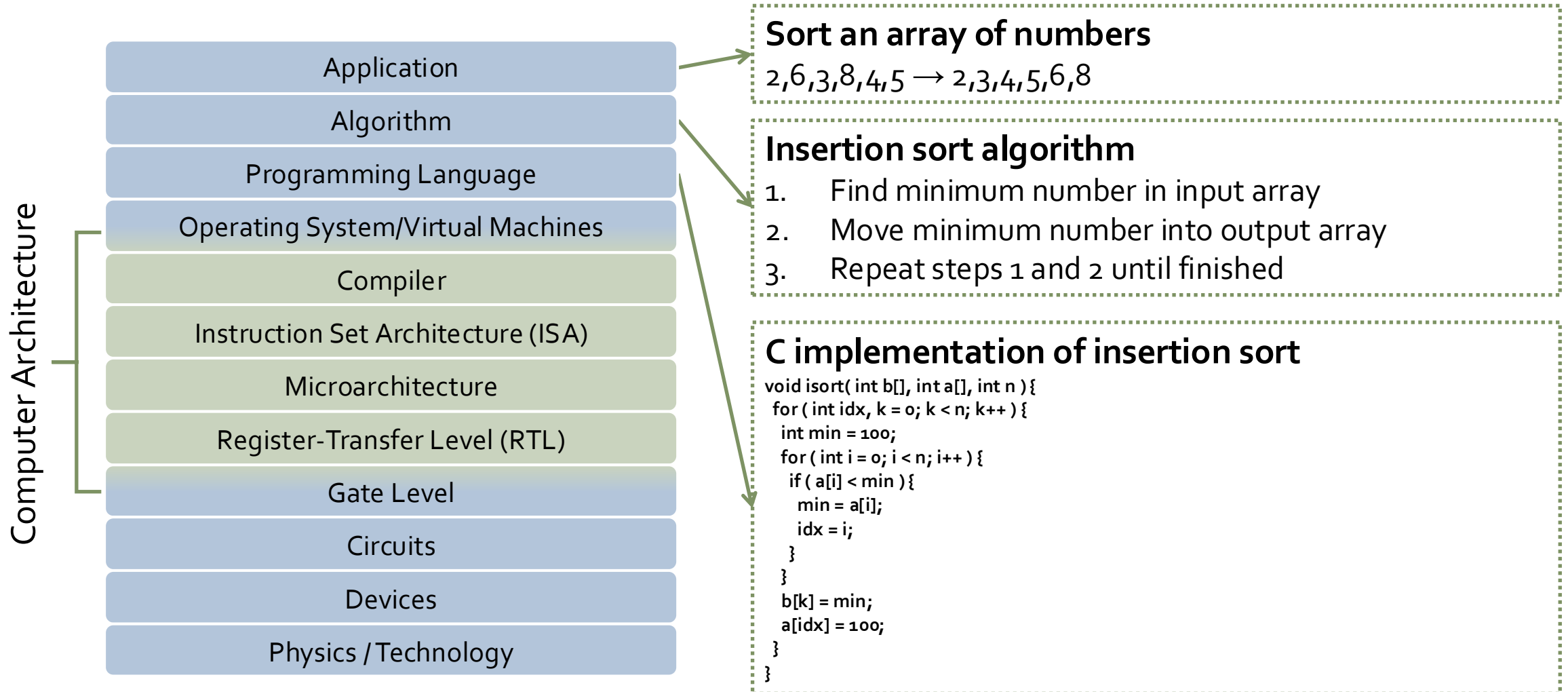
- What is Computer Architecture?
- Course Logistics

What is Computer Architecture?

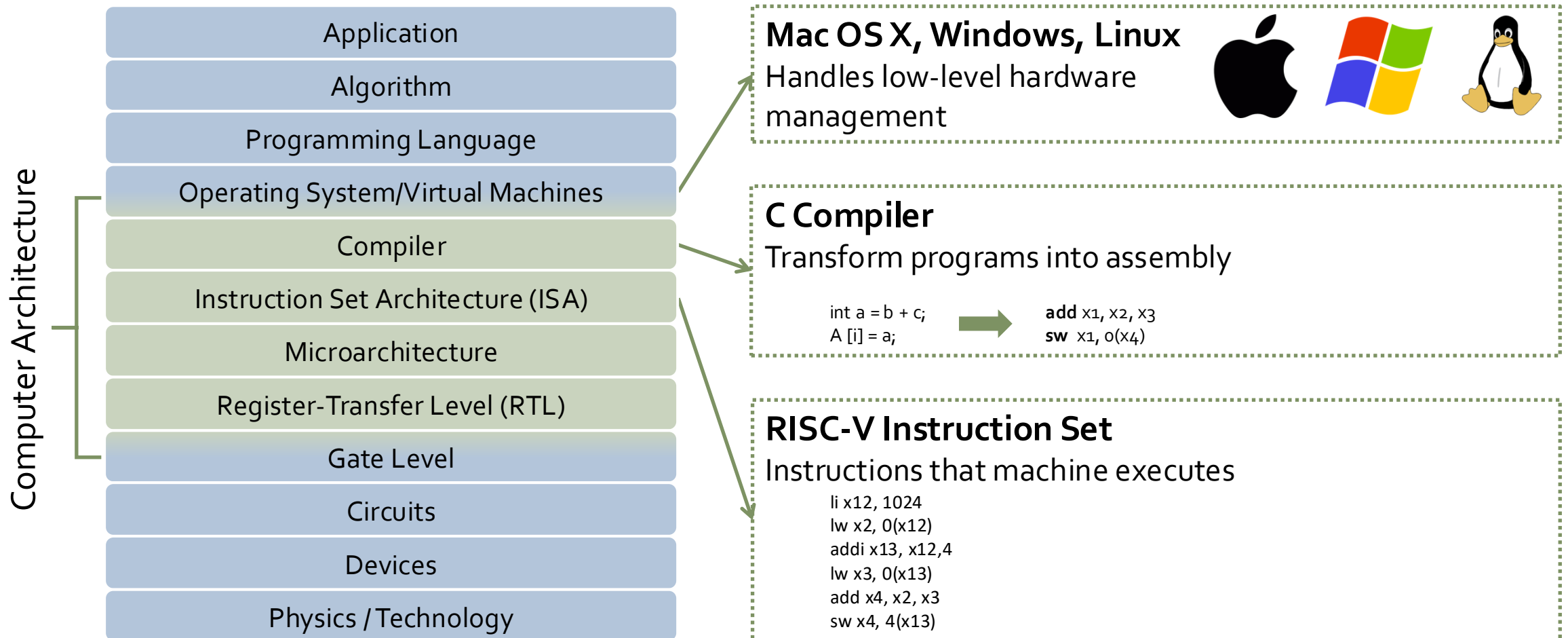


In its broadest definition, computer architecture is the **design of the abstraction/implementation layers** that allow us to execute information processing **applications** efficiently using manufacturing **technologies**.

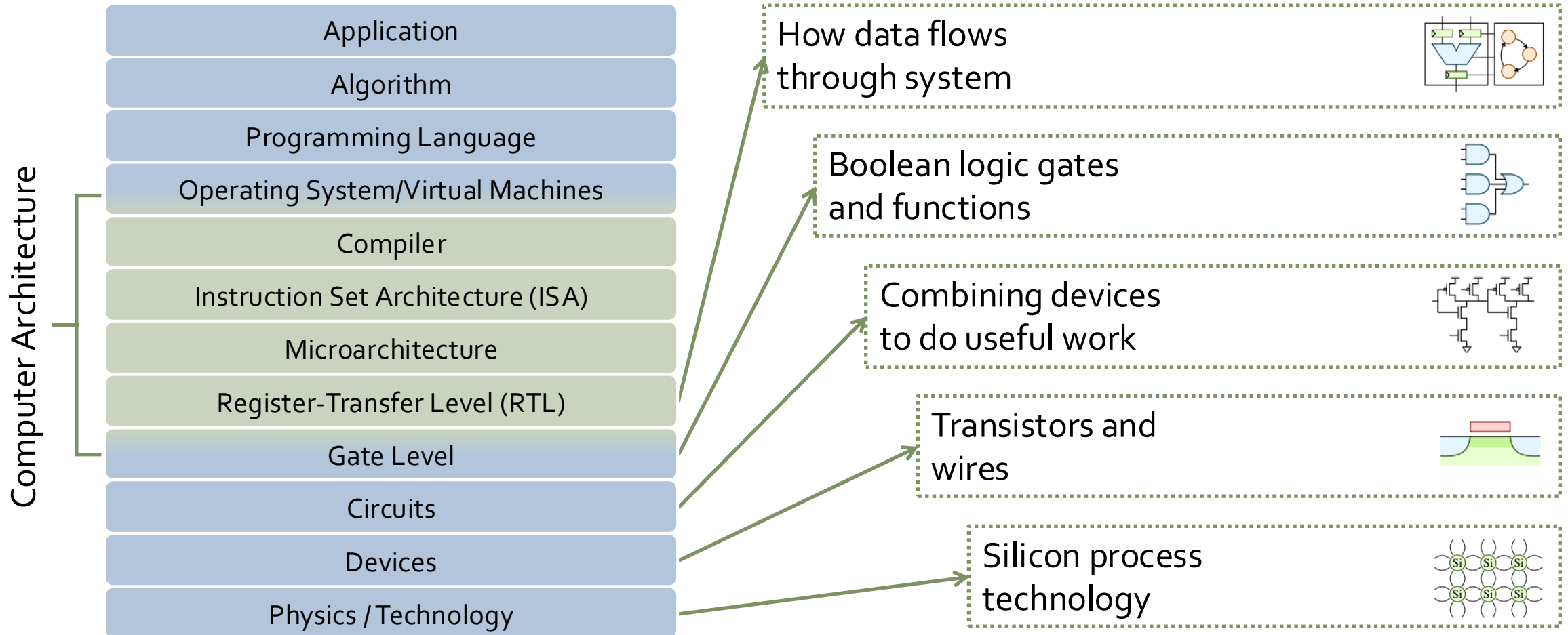
The Computer Systems Stack



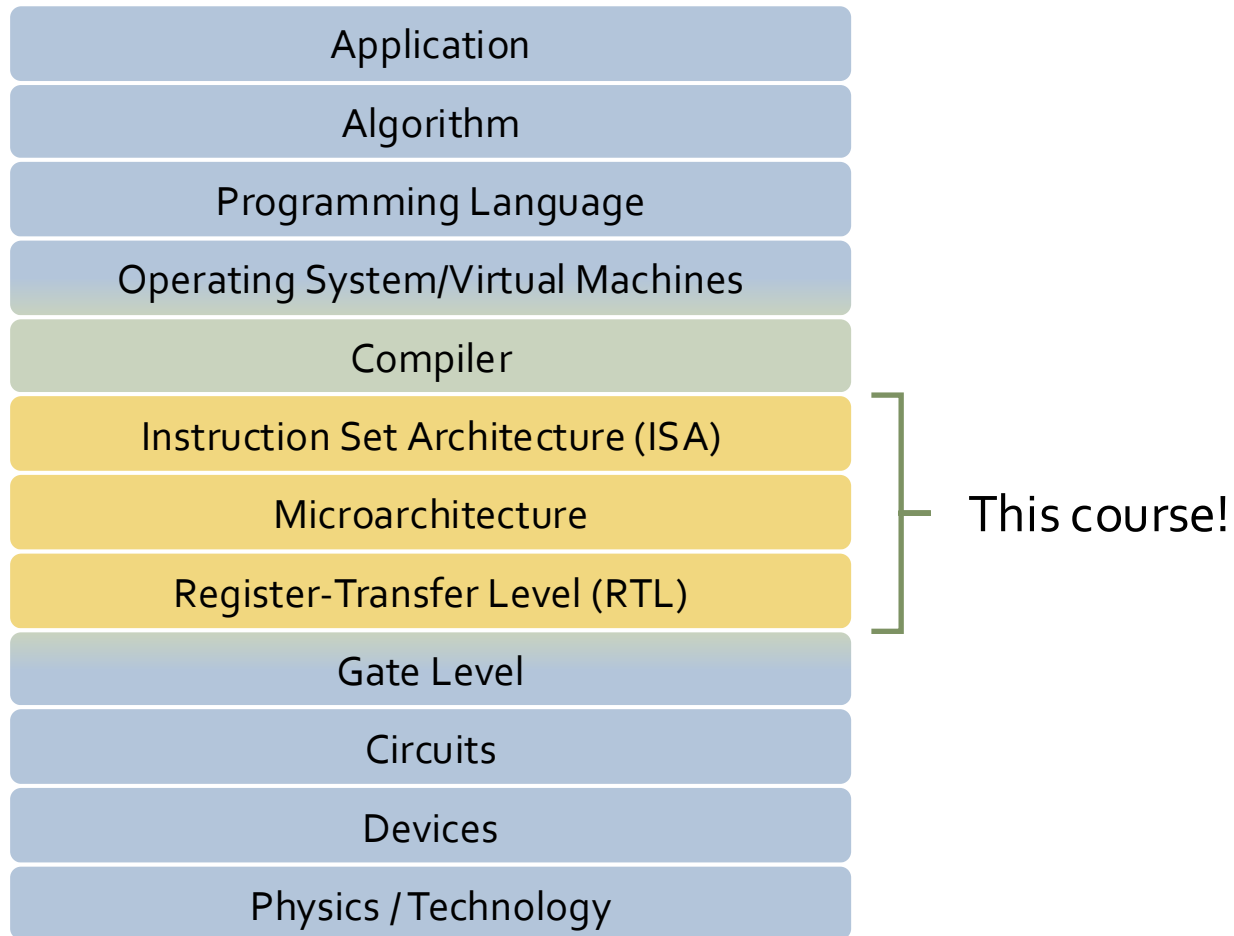
The Computer Systems Stack



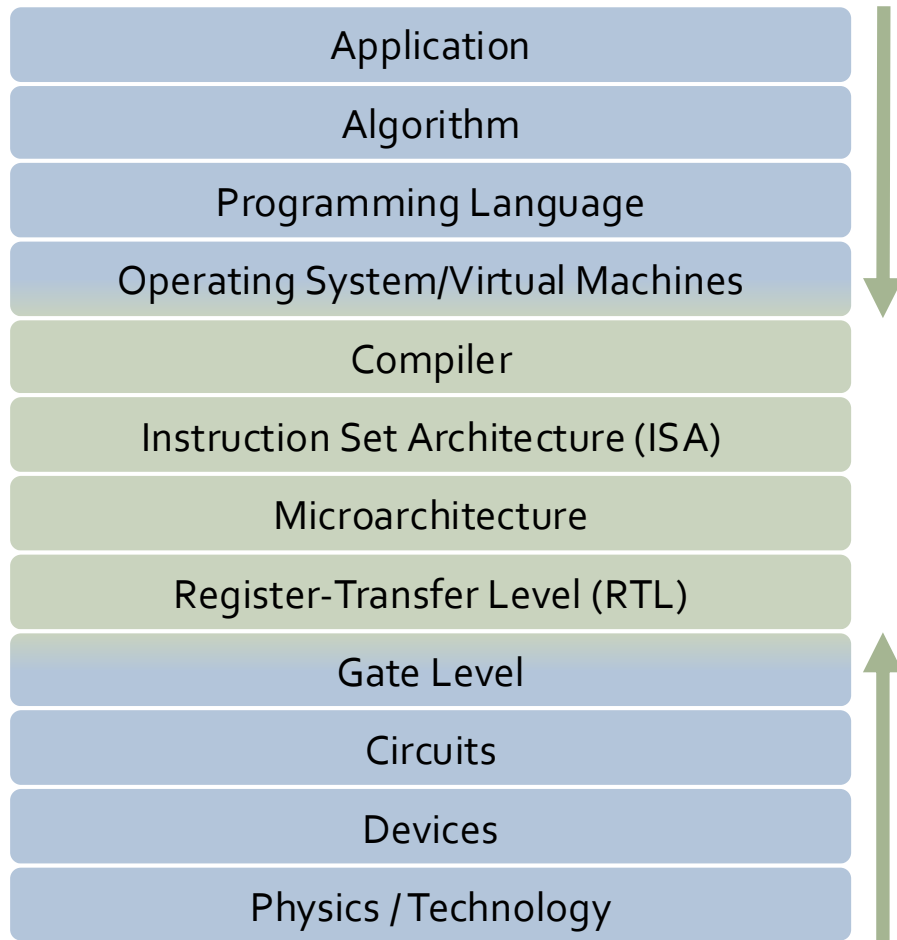
The Computer Systems Stack



The Computer Systems Stack



Architecture is Constantly Changing



Application Requirements:

- Suggest how to improve architecture
- Provide revenue to fund development

Architecture provides feedback to guide application and technology research directions

Technology Constraints:

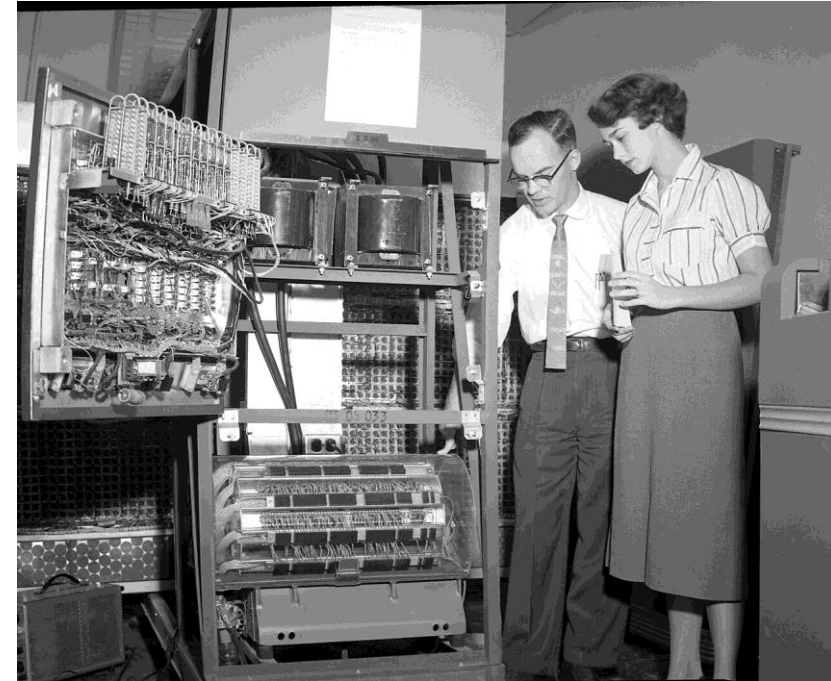
- Restrict what can be done efficiently
- New technologies make new arch possible

Computers Then...



IBM 650, 1962, NCTU

- The first mass-produced computer
- Almost 2,000 produced



[Cushing Memorial Library and Archives, Texas A&M,
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Computers Then...



Computers Now...



Personal



Autonomous



Embodied

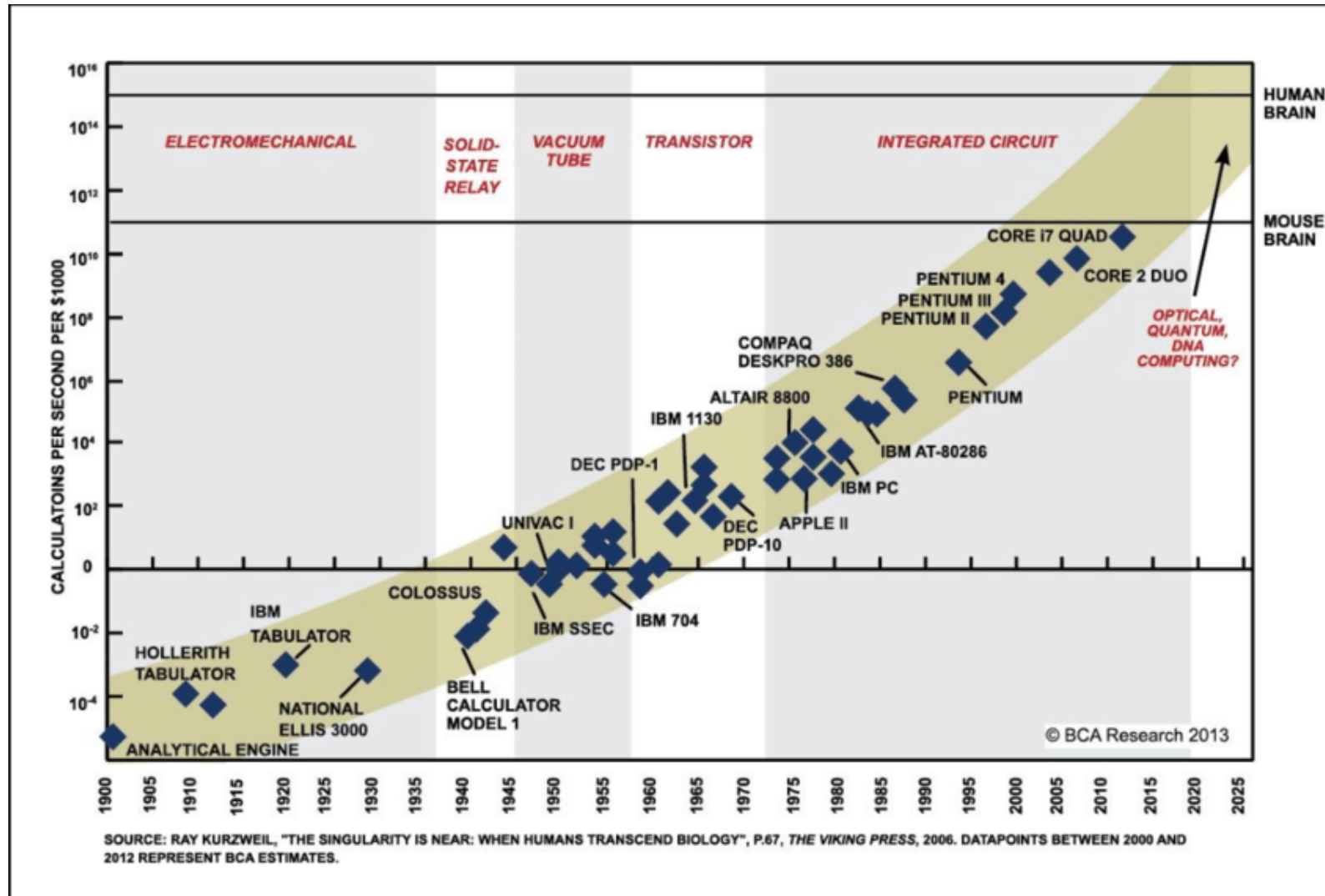


Spatial

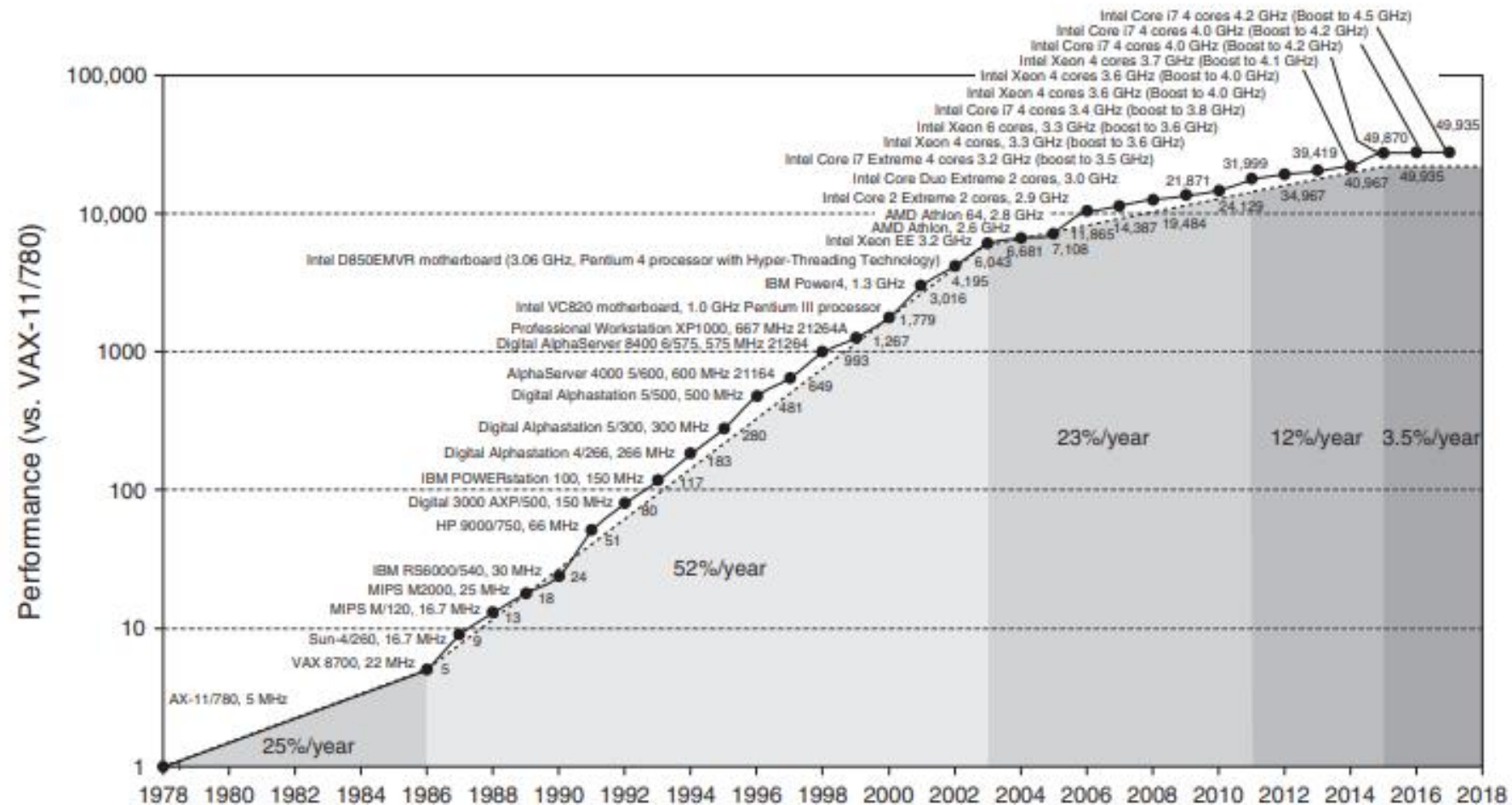


AI Factory

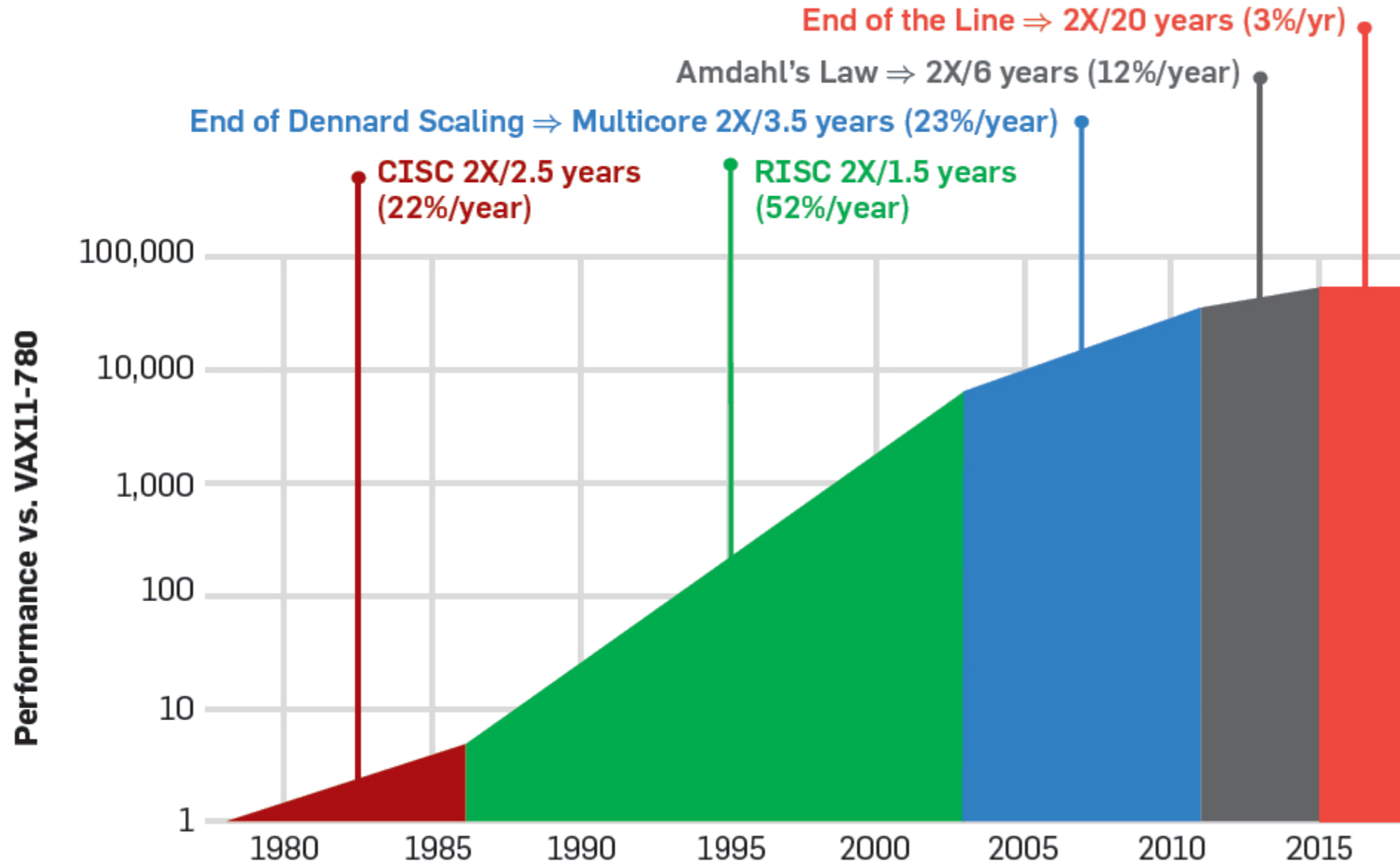
Moore's Law



Sequential Processor Performance

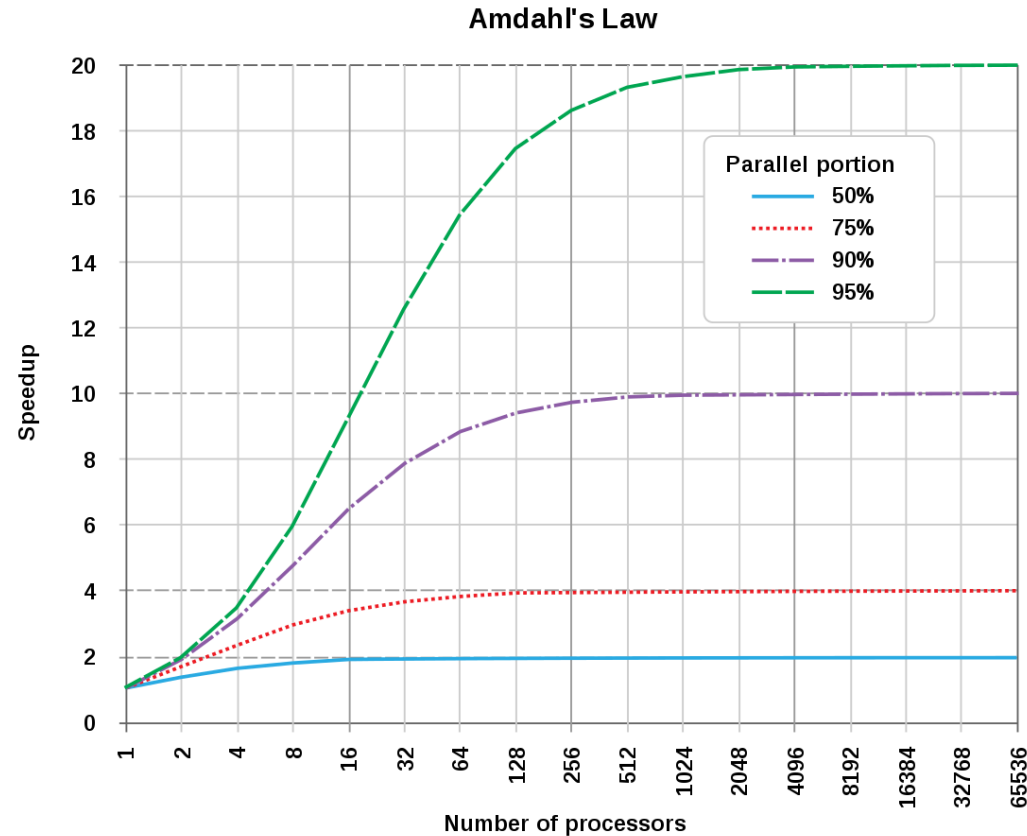


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Amdahl's Law

- $\text{Speedup} = \frac{\text{Execution time for entire task without using the enhancement}}{\text{Execution time for entire task using the enhancement when possible}}$



Upheaval in Computer Design

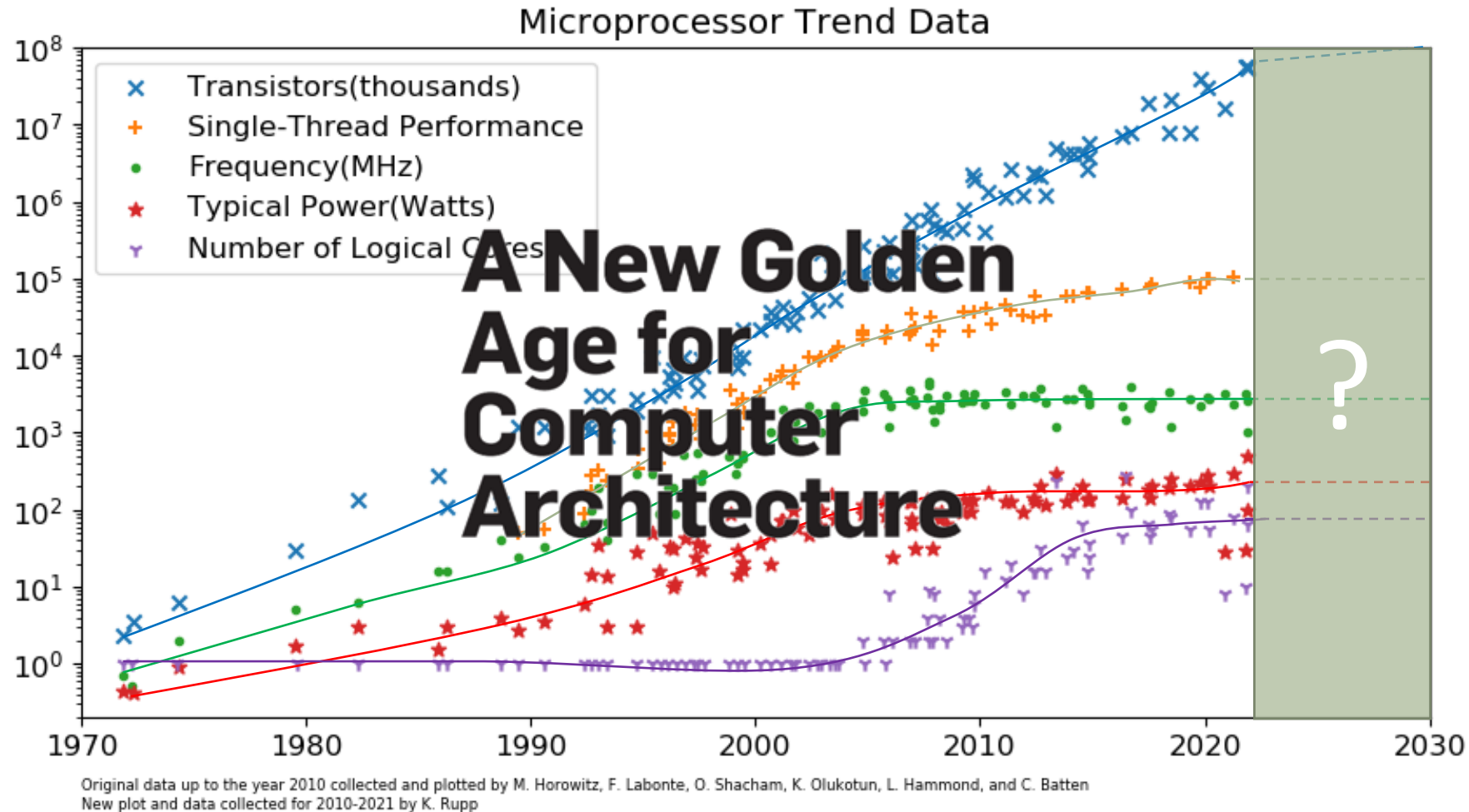
- Most of last 50 years, Moore's Law ruled
 - Scaling improved performance/energy without changing software model
- Last decade, technology scaling slowed/stopped
 - Dennard scaling is over (supply voltage ~fixed)
 - Moore's Law (cost/transistor) over?
 - No competitive replacement for CMOS in anytime soon
 - Energy efficiency is the main limiter
- 2020s shift
 - AI/ML drives compute demand
- No "free lunch" for software developers, must consider
 - Parallelism and heterogeneity are mandatory

Today's Dominant Target Systems

- Mobile (smartphone/tablet)
 - >1 billion sold/year
 - Dominated by ARM ISA in SoCs
 - Ship with AI/Neural engines + accelerators (vision, audio, security, sensors)
- Warehouse-Scale Computers (WSCs)
 - 100k+ cores per warehouse, cloud datacenters
 - x86 + growing Arm CPUs + custom accelerators
 - Energy & carbon footprint now a key bottleneck
- Embedded and Edge
 - Consumer electronics, automotive, IoT
 - Strong RISC-V growth in microcontrollers & AI edge chips
 - Edge AI (TinyML, on-device generative AI) expanding rapidly

Beyond Moore's Law?

Number of accelerators?
Parallelization and specialization?

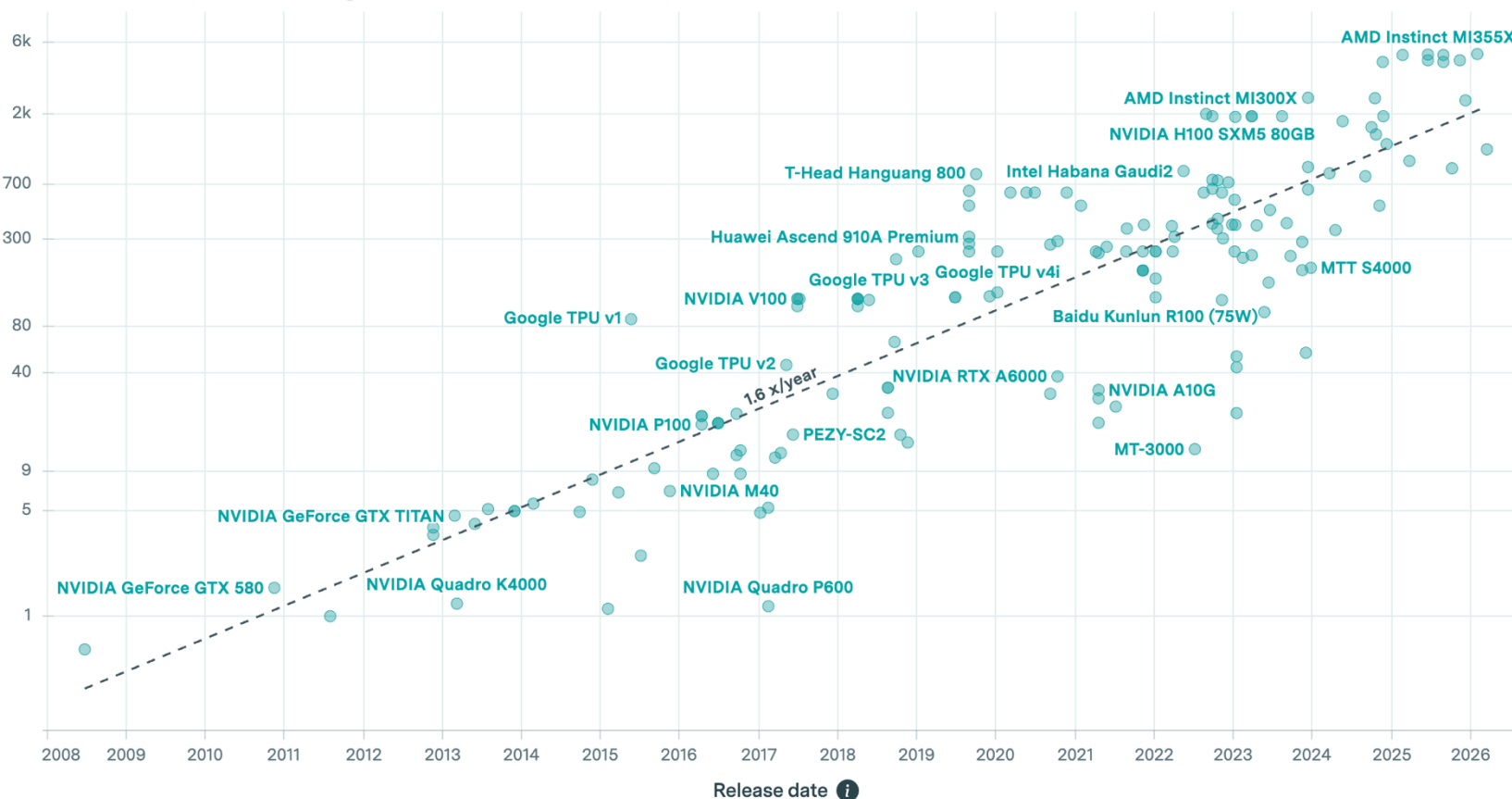


Trends in Machine Learning Hardware?

Machine Learning Hardware

Machine learning performance (TOP/s) ⓘ

161 Results ⓘ



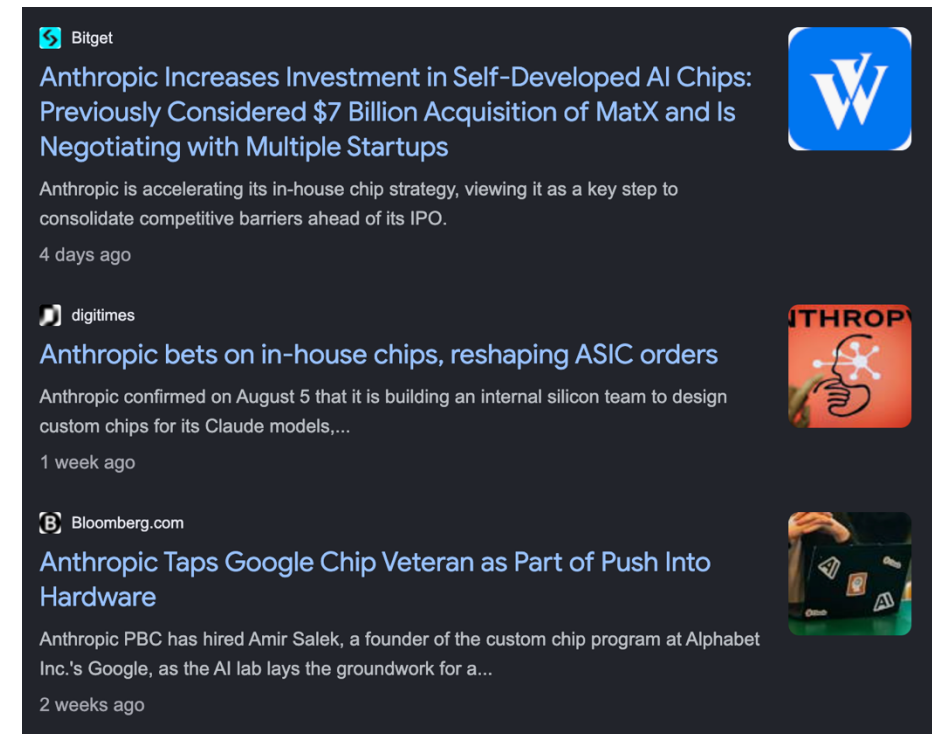
2008

2026

[Image Credit: Epoch AI]

The Verticalization of Silicon

- Shift: buying chips → building differentiated silicon
 - Apple: M-series SoCs
 - AWS, Google, Microsoft: custom CPUs + AI accelerators
 - Meta: custom AI accelerators
 - OpenAI: custom inference silicon
- Why?
 - Performance / watt
 - Lower system cost at scale
 - Less dependence on external vendors
 - Enables HW/SW co-design



Agenda

- What is Computer Architecture?
- Course Logistics

Course Logistics

- Instructor: Ting-Jung Chang (tingchang@cs.nycu.edu.tw)
 - Office Hours: T 16:00-17:00@EC707 by appointment
- Lectures: Tuesday 13:20 – 15:10@EDB27
- Text: Computer Architecture: A Quantitative Approach **6th Edition**
- Prerequisite: Computer Organization, or equivalent
- Course Webpage: e3

TAs

- 余承恩(johnyu.cs14@nycu.edu.tw)
 - 王少甫(tommywang.cs14@nycu.edu.tw)
 - 陳燁(hc.cs15@nycu.edu.tw)
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- Please email in advance
 - Check e3 for final updates (office hours and other info)

Course Format

- In-person lecture (2 hours)
- Independent study
 - Assigned textbook sections, papers, or keynotes
 - Short reading guides / study questions
 - No submission required
 - Assigned material is within exam scope

Course Structure

- Midterm (20%) (~10/27)
- Final (30%) (~12/22)
- Labs (50%)
 - 4 design labs (Verilog)
- Ungraded Problem Sets (0%)
 - Intended to help you learn the material
 - Feel free to discuss with other students and instructors
 - Useful for exam preparation

Preview of Lab Assignments

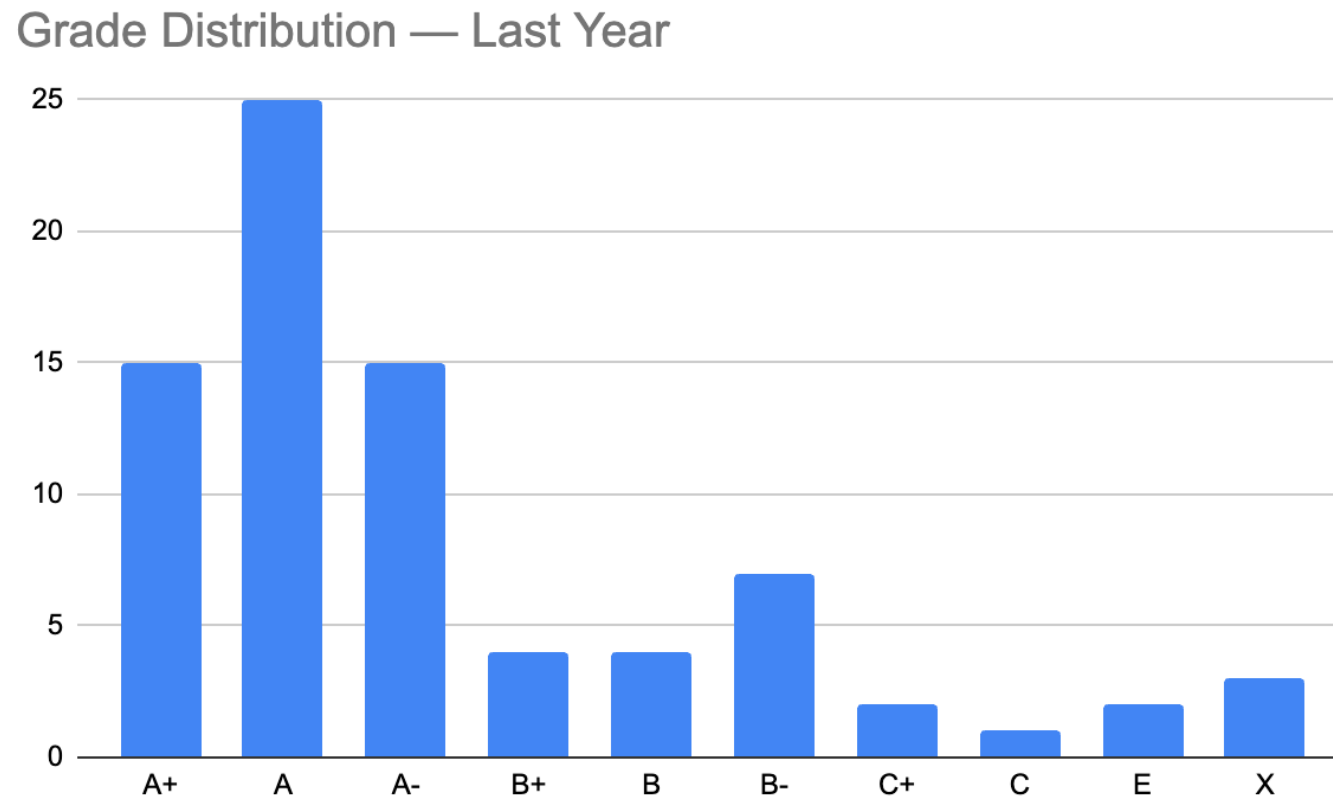
	Topic	Weight
Lab 0	Environment Setup	—
Lab 1	Pipelined CPU with Bypassing	5%
Lab 2	Reorder Buffer & Out-of-Order Writeback	5%
Lab 3	Superscalar Processor	20%
Lab 4	Cache Design	20%

Lab Integrity & Responsible AI Use

- Labs are designed to develop your own problem-solving ability. AI may support your learning, but it must not replace it.
- Be careful on AI usage
 - Explore ideas or debug with AI
 - But you must check, adapt, and make it your own
 - Work that lacks originality — AI or otherwise — will be penalized
- **No public sharing** of labs (GitHub, Google Drive, etc.)
- Integrity matters: violations can result in penalties, including failing the course

Statistics

- 78 students last till the end of the semester



Enrollment Cap Selection Process

- You must attend the first lecture and fill in the form



- Results will be out before the class next week

Computer Organization

- RISC-I (1982) fabbed in 5 μm NMOS, with a die area of 77 mm^2 , ran at 1 MHz. This chip is probably the first VLSI RISC
- Basic pipelined processor
- ~50,000 transistors

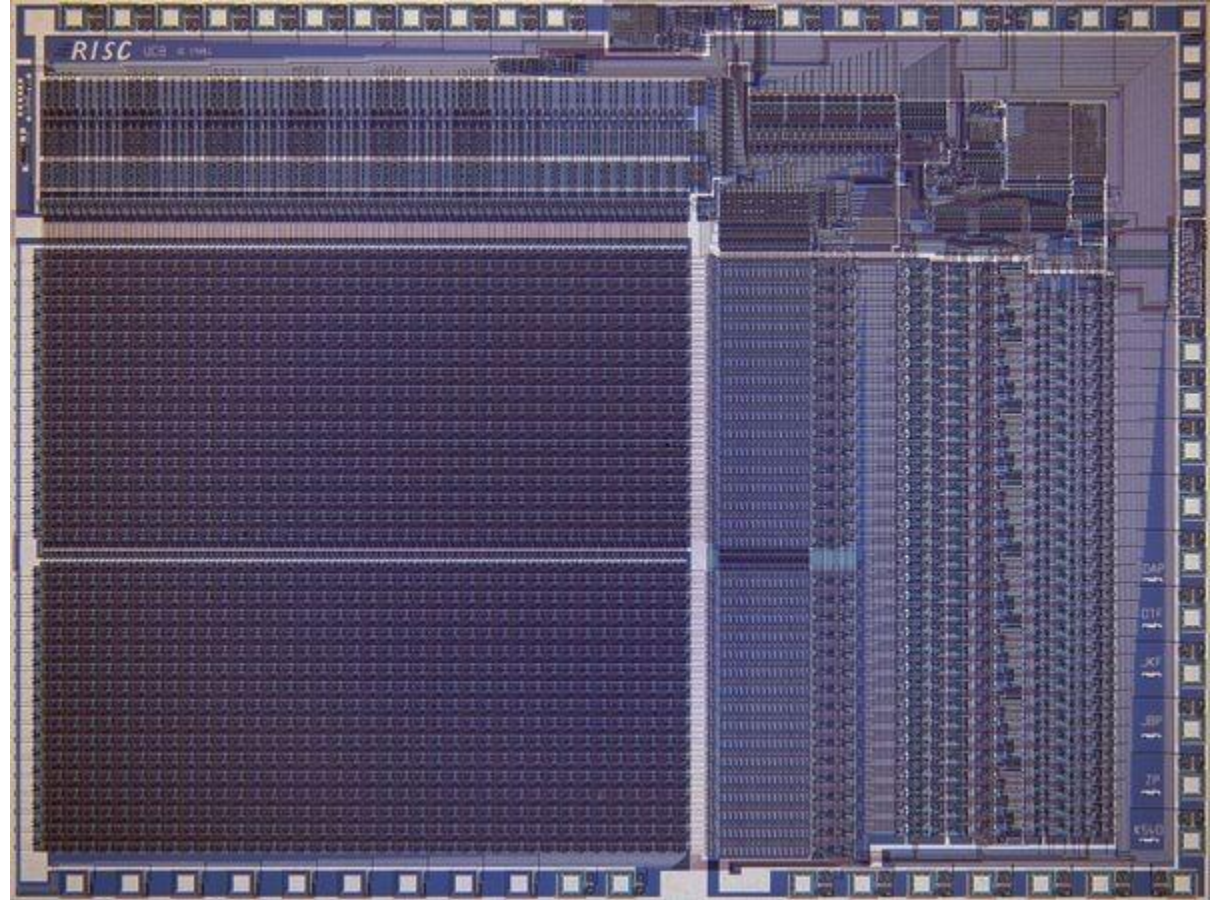


Photo of Berkeley RISC I, © University of California (Berkeley)

Computer Architecture

~731,000,000 transistors

- Instruction Level Parallelism
 - Superscalar
 - Very Long Instruction Word (VLIW)
- Long Pipelines (Pipeline Parallelism)
- Advanced Memory and Caches
- Data Level Parallelism
 - Vector
 - GPU
- Thread Level Parallelism
 - Multithreading
 - Multiprocessor
 - Multicore
 - Manycore
- More...

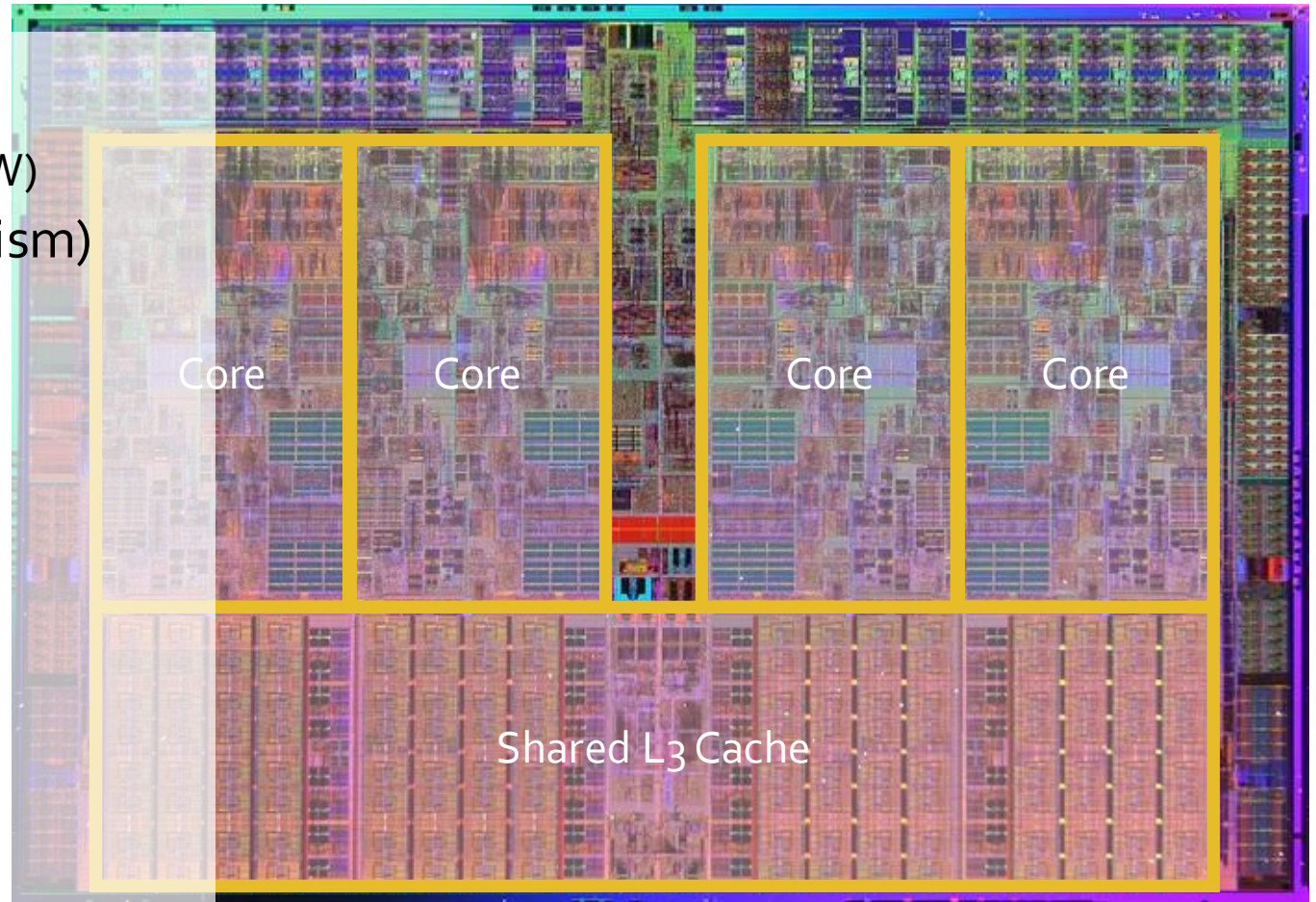


Photo of Intel Nehalem Processor, Original Core i7, © Intel 31

Course Logistics

- Your goal today: decide if you're coming back
- Notices (all materials available on e3)
 - Labo released
 - Verilog tips as supplementary material

Acknowledgements

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